

# Controlling DC Motors Through Raspberry Pi



In the last issue of EFY, we discussed how to use general-purpose inputs/outputs (GPIOs) available on Raspberry Pi for basic input/output operations. Let us now see how we can control DC motors through Raspberry Pi, with the help of three projects

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In the first project, we use the basic PHP-GPIO libraries and control two traction motors of a typical robot. In the second project, we move one step ahead and use WiringPi library to generate a pulse-width modulated (PWM) signal at one of the GPIOs. This PWM concept will be used in the third project to control the speed of DC motors.

## Project 1: Raspberry Pi controlling a dual H-Bridge

We are going to control two traction motors of a typical robot. One traction motor will make the robot walk in forward and reverse directions, while the other will be used to make hand movements—front and back. Overall, we will be controlling two coreless DC motors, as indicated in Table I, using the keyboard of a Wi-Fi connected laptop.

## Circuit and working

Fig. 2 shows the circuit setup for project 1. A dual H-Bridge driver L293D (IC1) is used to drive motors M1 and M2. It is capable of supplying 600mA current per channel at 12 V.

Input pins 2, 7, 10 and 15 of IC1 are interfaced to Raspberry Pi GPIO connector as shown in Fig. 2. Pin 8 is for motor power supply. It is directly connected to on-board 5V power supply of Raspberry Pi available at pin 2. Pins 1 and 9 are used to enable motors M1 and M2, respectively. The speed of motors can be controlled by supplying a PWM pulse at these pins. But since we are not controlling the speed here,

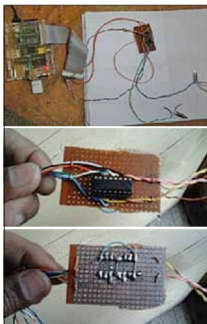


Fig. 1: Author's prototype

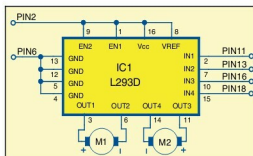


Fig. 2: Circuit for project 1

we have directly connected these two pins to 5V supply pin. Ground pins 4, 5, 12 and 13 of IC1 are connected to pin 6 (ground) of Raspberry Pi.

Working of the circuit is simple. Once you run the program (explained in software section), you will get an output as shown in Fig. 3. Now you

TABLE I  
Input Choices and Corresponding Motor Movements

| Enter choice | Motion           |
|--------------|------------------|
| 1            | M1 forward start |
| 11           | M1 forward stop  |
| 2            | M1 reverse start |
| 21           | M1 reverse stop  |
| 3            | M2 forward start |
| 31           | M2 forward stop  |
| 4            | M2 reverse start |
| 41           | M2 reverse stop  |
| 5            | Both motors stop |

have to enter choices from your keyboard. The motors will move or stop accordingly as shown in Table I.

## Software

The software for this project is designed in PHP. To run this program successfully, you need two things: Raspberry Pi and PHP-GPIO library.

First, access Raspberry Pi remotely so that you don't have to connect a separate monitor, keyboard and mouse to Raspberry Pi. You can refer the article 'Remote Access With Raspberry Pi' published in June issue to learn various ways of accessing it from your existing computer. Once you have accessed Raspberry Pi remotely, PHP can easily be installed in the terminal by the following command:

```
sudo apt-get install php5
```

This command will run without any issue and eventually come back